UMSA - INFORMÁTICA - BASE DE DATOS II - INF252
AUX: UNIV. MOISES MARTIN CONDORI YUJRA

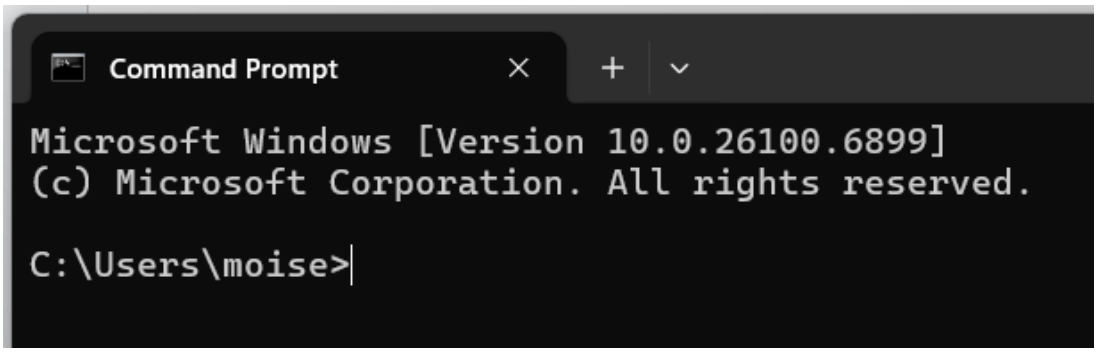
Guía Completa: Exportación e Importación de Bases de Datos por Consola (Data Pump)

Esta práctica te guiará paso a paso para respaldar una base de datos existente y clonarla en un usuario completamente nuevo utilizando exclusivamente la línea de comandos de Windows y Oracle (Data Pump).

Fase 1: Preparación del Entorno y Usuarios

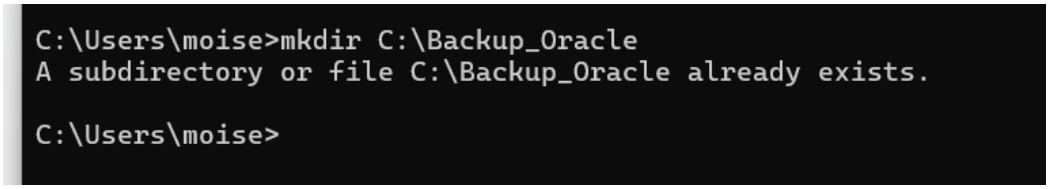
Antes de mover datos, debemos crear una carpeta física en la computadora, enseñarle a Oracle dónde está y preparar al nuevo usuario que recibirá la información.

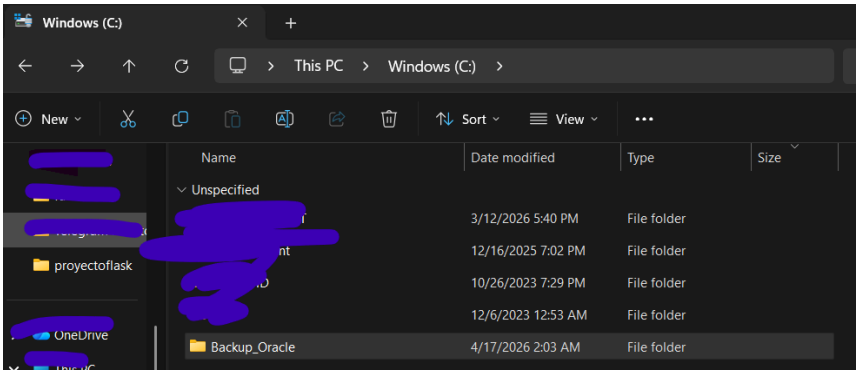
1. Abre la consola de Windows (Símbolo del sistema). Presiona `Windows + R`, escribe **`cmd`** y presiona Enter.



2. Crea la carpeta de respaldo en tu disco duro:

DOS
mkdir C:\Backup_Oracle





3. Ingresa a Oracle con privilegios de administrador supremo(ingresar password de SYSTEM O SYSDBA):

sqlplus sys as sysdba

```
C:\Users\moise>sqlplus sys as sysdba

SQL*Plus: Release 21.0.0.0.0 - Production on Fri Apr 17 01:54:51 2026
Version 21.3.0.0.0

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Enter password:

Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL>
```

4. Conéctate al contenedor correcto donde viven nuestras bases de datos:

ALTER SESSION SET CONTAINER = XEPDB1;

```
SQL> ALTER SESSION SET CONTAINER = XEPDB1;

Session altered.

SQL>
```

5. Crea el "Directorio Lógico" (el puente entre Oracle y tu carpeta de Windows):
SQL
CREATE OR REPLACE DIRECTORY DIR_BACKUP AS 'C:\Backup_Oracle';

```
SQL> CREATE OR REPLACE DIRECTORY DIR_BACKUP AS 'C:\Backup_Oracle';

Directory created.

SQL>
```

6. Crea el usuario nuevo que recibirá los datos (nuestro clon) y asígnale su contraseña:
SQL
CREATE USER BD_TESTCMD2 IDENTIFIED BY admin123;

```
SQL> CREATE USER BD_TESTCMD2 IDENTIFIED BY admin123;

User created.

SQL>
```

7. Otorga los permisos necesarios para que el nuevo usuario pueda conectarse, crear tablas y leer la carpeta de respaldos:
SQL
GRANT CONNECT, RESOURCE TO BD_TESTCMD2;
ALTER USER BD_TESTCMD2 QUOTA UNLIMITED ON USERS;
GRANT READ, WRITE ON DIRECTORY DIR_BACKUP TO BD_TESTCMD2;

```
SQL> GRANT CONNECT, RESOURCE TO BD_TESTCMD2;

Grant succeeded.

SQL> ALTER USER BD_TESTCMD2 QUOTA UNLIMITED ON USERS;

User altered.

SQL> GRANT READ, WRITE ON DIRECTORY DIR_BACKUP TO BD_TESTCMD2;

Grant succeeded.

SQL>
```

8. Otorga permiso de lectura/escritura en la carpeta al usuario original que vamos a exportar:
SQL
GRANT READ, WRITE ON DIRECTORY DIR_BACKUP TO BD2;

```
SQL> GRANT READ, WRITE ON DIRECTORY DIR_BACKUP TO BD2;

Grant succeeded.

SQL> |
```

EXIT

Sal de la consola de Oracle para volver a Windows:
SQL
EXIT

```
SQL> exit
Disconnected from Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

C:\Users\moise>
```

Fase 2: Verificación de la Base de Datos Origen

Como buena práctica, siempre debemos confirmar que el usuario de origen tiene las tablas que queremos respaldar.

1. Conéctate al usuario original que contiene los datos:
   DOS
   sqlplus BD2/admin123@localhost:1521/XEPDB1

```
C:\Users\moise>sqlplus BD2/admin123@localhost:1521/XEPDB1

SQL*Plus: Release 21.0.0.0.0 - Production on Fri Apr 17 02:00:33 2026
Version 21.3.0.0.0

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Last Successful login time: Wed Apr 15 2026 19:37:52 -04:00

Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL>
```

2. Lista las tablas existentes para confirmar:
   SQL
   SELECT table_name FROM user_tables;

```
SQL> SELECT table_name FROM user_tables;

TABLE_NAME
--------------------------------------------------------------------------------
DOCENTE
AULA
MATERIA
ESTUDIANTE
INSCRIPCION
HORARIO
PAGO
EVALUACION
ASISTENCIA
MATERIAL_DIDACTICO
VEHICULOS

11 rows selected.

SQL>
```

(Debes ver la lista de tus tablas: DOCENTE, AULA, MATERIA, ESTUDIANTE, etc.)

3. Sal de la consola de Oracle:
   SQL
   EXIT

```
SQL> EXIT
Disconnected from Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

C:\Users\moise>
```

Fase 3: La Exportación (Data Pump)

Ahora empaquetaremos todas esas tablas en un archivo `.dmp`.

1. En la consola de Windows, ejecuta el comando de exportación:
   DOS
   expdp BD2/admin123@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Export_BD2.log SCHEMAS=BD2

```
C:\Users\moise>expdp BD2/admin123@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Export_BD2.log SCHEMAS=BD2

Export: Release 21.0.0.0.0 - Production on Fri Apr 17 02:02:53 2026
Version 21.3.0.0.0

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Connected to: Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Starting "BD2"."SYS_EXPORT_SCHEMA_01":  BD2/*******@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Export_BD2.log SCHEMAS=
BD2
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/STATISTICS/MARKER
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
. . exported "BD2"."HORARIO"                              7.804 KB      20 rows
. . exported "BD2"."DOCENTE"                              7.679 KB      20 rows
. . exported "BD2"."MATERIAL_DIDACTICO"                   7.546 KB      20 rows
. . exported "BD2"."EVALUACION"                           7.507 KB      20 rows
. . exported "BD2"."ESTUDIANTE"                           7.382 KB      20 rows
. . exported "BD2"."VEHICULOS"                            7.492 KB       7 rows
. . exported "BD2"."PAGO"                                 7.414 KB      20 rows
. . exported "BD2"."ASISTENCIA"                           7.328 KB      20 rows
. . exported "BD2"."INSCRIPCION"                          7.281 KB      20 rows
. . exported "BD2"."MATERIA"                              6.921 KB      20 rows
. . exported "BD2"."AULA"                                 6.820 KB      20 rows
Master table "BD2"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
******************************************************************************
```
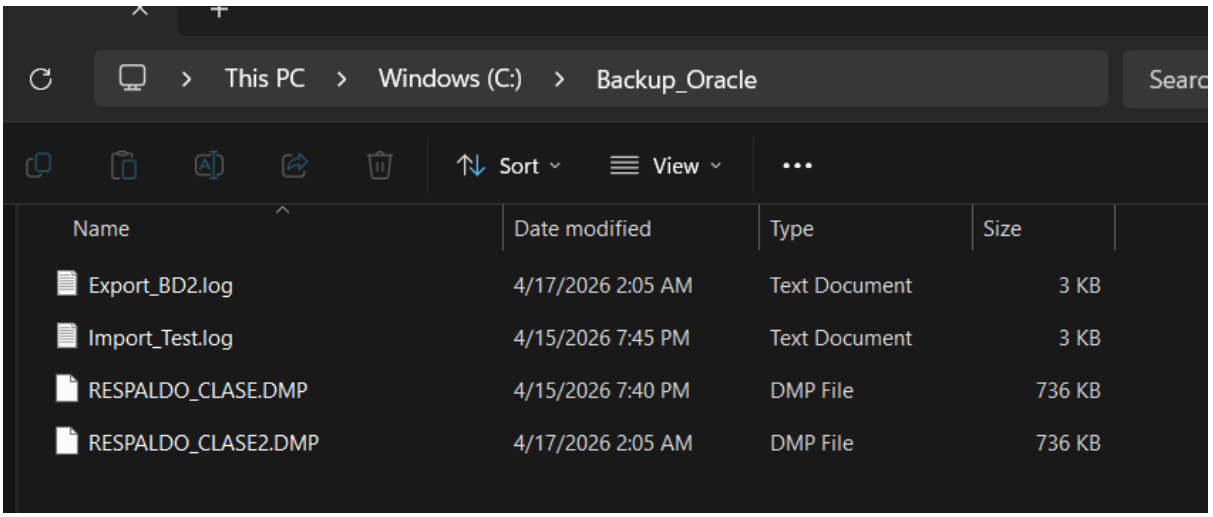

```
******************************************************************************
Dump file set for BD2.SYS_EXPORT_SCHEMA_01 is:
  C:\BACKUP_ORACLE\RESPALDO_CLASE2.DMP
Job "BD2"."SYS_EXPORT_SCHEMA_01" successfully completed at Fri Apr 17 02:05:18 2026 elapsed 0 00:02:15


C:\Users\moise>
```

2. Espera a que el proceso termine. Al finalizar, verás el mensaje `"successfully completed"` y el archivo `Respaldo_Clase2.dmp` estará guardado en tu disco C:.



Fase 4: Verificación del Usuario Destino (Pre-Importación)

Vamos a confirmar que el usuario nuevo está completamente vacío antes de inyectarle los datos.

1. Conéctate al usuario nuevo:
   DOS
   sqlplus BD_TESTCMD2/admin123@localhost:1521/XEPDB1

```
C:\Users\moise>sqlplus BD_TESTCMD2/admin123@localhost:1521/XEPDB1

SQL*Plus: Release 21.0.0.0.0 - Production on Fri Apr 17 02:12:34 2026
Version 21.3.0.0.0

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Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL>
```

2. Consulta sus tablas:
   SQL
   SELECT table_name FROM user_tables;

   (La consola debe responder `no rows selected`, confirmando que está vacío).

```
SQL> SELECT table_name FROM user_tables;

no rows selected

SQL>
```

3. Sal de la consola de Oracle:
   SQL
   EXIT

```
SQL> EXIT
Disconnected from Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

C:\Users\moise>
```

Fase 5: La Importación (Data Pump)

Llegó el momento de clonar los datos hacia el usuario nuevo utilizando el parámetro de redirección.

1. En la consola de Windows, ejecuta el comando de importación:
   DOS
   impdp BD_TESTCMD2/admin123@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Import_Test2.log REMAP_SCHEMA=BD2:BD_TESTCMD2

   (Nota importante: El comando `REMAP_SCHEMA` es fundamental, ya que traduce la propiedad de las tablas del usuario BD2 al nuevo usuario BD_TESTCMD2 de forma automática).

```
C:\Users\moise>impdp BD_TESTCMD2/admin123@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Import_Test2.log REMAP_SCHEMA=BD2:
BD_TESTCMD2

Import: Release 21.0.0.0.0 - Production on Fri Apr 17 02:14:49 2026
Version 21.3.0.0.0

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Connected to: Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Master table "BD_TESTCMD2"."SYS_IMPORT_FULL_01" successfully loaded/unloaded
Starting "BD_TESTCMD2"."SYS_IMPORT_FULL_01":  BD_TESTCMD2/********@localhost:1521/XEPDB1 DIRECTORY=DIR_BACKUP DUMPFILE=Respaldo_Clase2.dmp LOGFILE=Import_Te
st2.log REMAP_SCHEMA=BD2:BD_TESTCMD2
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
. . imported "BD_TESTCMD2"."HORARIO"                     7.804 KB      20 rows
. . imported "BD_TESTCMD2"."DOCENTE"                     7.679 KB      20 rows
. . imported "BD_TESTCMD2"."MATERIAL_DIDACTICO"          7.546 KB      20 rows
. . imported "BD_TESTCMD2"."EVALUACION"                  7.507 KB      20 rows
. . imported "BD_TESTCMD2"."ESTUDIANTE"                  7.382 KB      20 rows
. . imported "BD_TESTCMD2"."VEHICULOS"                   7.492 KB       7 rows
. . imported "BD_TESTCMD2"."PAGO"                        7.414 KB      20 rows
. . imported "BD_TESTCMD2"."ASISTENCIA"                  7.328 KB      20 rows
. . imported "BD_TESTCMD2"."INSCRIPCION"                 7.281 KB      20 rows
. . imported "BD_TESTCMD2"."MATERIA"                     6.921 KB      20 rows
. . imported "BD_TESTCMD2"."AULA"                        6.820 KB      20 rows
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/STATISTICS/MARKER
Job "BD_TESTCMD2"."SYS_IMPORT_FULL_01" successfully completed at Fri Apr 17 02:16:04 2026 elapsed 0 00:01:08
```

```
Processing object type SCHEMA_EXPORT/STATISTICS/MARKER
Job "BD_TESTCMD2"."SYS_IMPORT_FULL_01" successfully completed at Fri Apr 17 02:16:04 2026 elapsed 0 00:01:08


C:\Users\moise>
```

Fase 6: Comprobación Final

Finalmente, verificaremos que los datos se clonaron correctamente con todos sus registros.

1. Conéctate por última vez al usuario nuevo:
   DOS
   sqlplus BD_TESTCMD2/admin123@localhost:1521/XEPDB1

```
C:\Users\moise>sqlplus BD_TESTCMD2/admin123@localhost:1521/XEPDB1

SQL*Plus: Release 21.0.0.0.0 - Production on Fri Apr 17 02:17:44 2026
Version 21.3.0.0.0

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Last Successful login time: Fri Apr 17 2026 02:14:49 -04:00

Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> |
```

2. Verifica que las tablas ya existen:
   SQL
   SELECT table_name FROM user_tables;

   (Ahora sí verás la lista de las 11 tablas importadas).

```
SQL> SELECT table_name FROM user_tables;

TABLE_NAME
--------------------------------------------------------------------------------
MATERIAL_DIDACTICO
ASISTENCIA
AULA
HORARIO
MATERIA
ESTUDIANTE
EVALUACION
DOCENTE
INSCRIPCION
VEHICULOS
PAGO

11 rows selected.

SQL> |
```

3. Realiza consultas de prueba para verificar los registros por dentro:
   SQL
   SELECT * FROM DOCENTE;

SELECT COUNT(*) FROM ESTUDIANTE;

   (La consola te mostrará la lista de docentes y confirmará que existen exactamente 20 estudiantes matriculados).

```
SQL> SELECT COUNT(*) FROM ESTUDIANTE;

  COUNT(*)
----------
        20

SQL>
```

```
SQL> SELECT * FROM DOCENTE;

DOCENTE_ID NOMBRE
---------- --------------------------------------------------
APELLIDO
--------------------------------------------------
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
         1 Carlos
Mendoza
Matemßticas
cmendoza@uni.edu


DOCENTE_ID NOMBRE
---------- --------------------------------------------------
APELLIDO
--------------------------------------------------
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
         2 Ana
Torres
Historia
atorres@uni.edu


DOCENTE_ID NOMBRE
---------- --------------------------------------------------
APELLIDO
--------------------------------------------------
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
         3 Luis
```

```
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
        18 Diana
Flores
Economϕa
dflores@uni.edu


DOCENTE_ID NOMBRE
---------- --------------------------------------------------
APELLIDO
--------------------------------------------------
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
        19 Vϕctor
Morales
Administraci≤n
vmorales@uni.edu


DOCENTE_ID NOMBRE
---------- --------------------------------------------------
APELLIDO
--------------------------------------------------
ESPECIALIDAD
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
        20 Rosa
Ortiz
Marketing
rortiz@uni.edu


20 rows selected.

SQL>
```